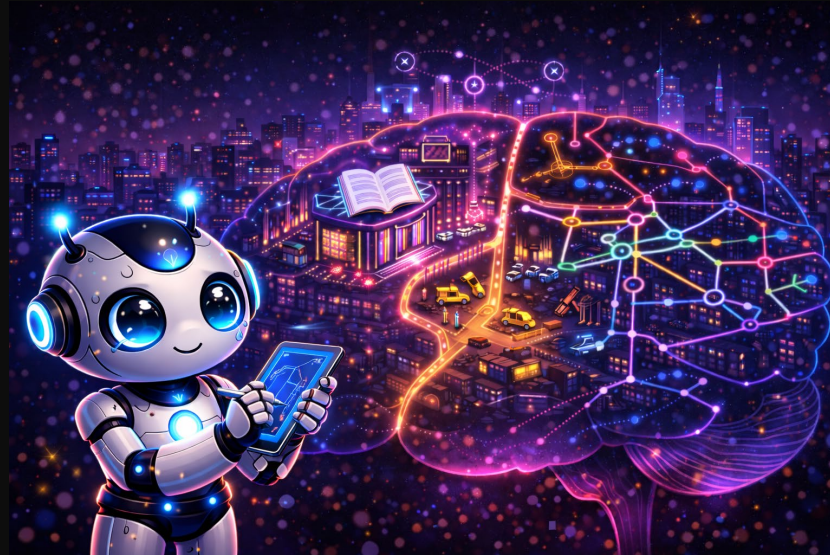




Module 06

Memory, Learning, and AI Architectures



From Remembering to Reasoning

Houston City College • AI & Robotics Program

Professor Patricia McManus



What You Will Learn

1. Explain the four biological memory systems and their brain structures
2. Describe hippocampal consolidation during sleep — and why it matters for AI
3. Explain LTP, LTD, and neuromodulators as the cellular machinery of learning
4. Connect hippocampal replay to experience replay in Deep RL
5. Compare AI memory architectures to their biological inspirations
6. Articulate the biological plausibility debate around backpropagation



Where We Are in the Course

Module 04



Module 05



Module 06



Module 07–08

CONNECT

**SENSE
DECIDE**

**REMEMBER
& LEARN**

*Once the brain perceives the world, how does it **STORE** information,
RETRIEVE it later, and **LEARN** from experience?*

The Man Who Couldn't Make Memories

Henry Molaison (1926–2008)

1953: Surgeons removed most of his hippocampus to treat severe epilepsy

Seizures improved — but he lost the ability to form new memories

Lived in a permanent present for 55 years. Met his doctors anew every day.

Key revelation: Memory is not one thing — it is a collection of separate systems.





PART 1

How the Brain Remembers

Conceptual — No Math Required



Four Memory Systems — Not All Memories Are Equal

Working Memory — Hold info right now — Prefrontal Cortex — AI: Context window

Episodic Memory — Personal experiences — Hippocampus — AI: RAG, replay buffers

Semantic Memory — Facts & concepts — Cortex (distributed) — AI: Model weights

Procedural Memory — Skills & habits — Basal ganglia — AI: RL policies

Most AI systems have 1–2 of these. The brain runs all four simultaneously.

Working Memory — The Brain's Scratchpad

Capacity: ~4 items (Cowan) to 7 ± 2 items (Miller, 1956)

Duration: Seconds — gone when you're distracted

Brain region: Prefrontal cortex (sustained firing)

🧠 → 🤖 AI: Context window in LLMs, hidden state in RNNs, CPU registers

Key gap: Biological WM is flexible (you focus more on hard tasks). AI context windows are fixed.



Episodic Memory: Your Personal Time Machine

What, where, when — including emotions and context

Coined by Endel Tulving (1972)

Hippocampus = index/binding agent

Links distributed cortical representations into unified episodes

Over time, consolidation strengthens cortical links → hippocampus lets go

H.M.'s devastation: Lost episodic memory completely, but kept semantic and procedural

🧠 → 🤖 AI: RAG systems, memory-augmented networks, episodic control in RL



Semantic & Procedural Memory

Semantic Memory — Facts & concepts without personal context

"Paris is in France" — you don't remember when you learned this

Brain: Distributed across cortex. Episodic → Semantic over time.

🧠 → 🤖 **AI = Trained model weights**

Procedural Memory — Skills & habits, largely unconscious

Riding a bike, typing, playing guitar

Brain: Basal ganglia + cerebellum

H.M. learned mirror tracing despite no episodic memory!

🧠 → 🤖 **AI = RL policies, robot controllers**



⚠ Memory Is NOT a Video Recorder

Memory is RECONSTRUCTIVE

Every recall actively rebuilds the memory from fragments — and it can change each time

This is why eyewitness testimony is unreliable
This is why you can 'remember' events that never happened

The tradeoff: AI has perfect recall but no flexibility.
Biology has flexible recall but introduces errors.



The Hippocampus The Brain's Memory Engine

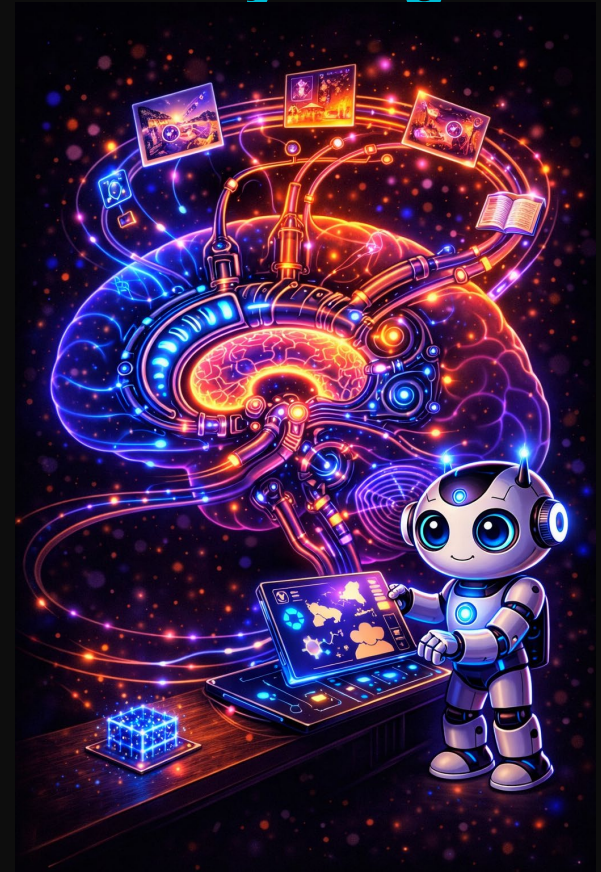
The Trisynaptic Circuit:

Entorhinal Cortex → Dentate Gyrus → CA3 → CA1

Dentate Gyrus: Pattern separation — making similar inputs distinct

CA3: Pattern completion — retrieving full memories from partial cues

CA1: Novelty detection — is this new or familiar?



Memory Consolidation — Why You Need Sleep to Learn

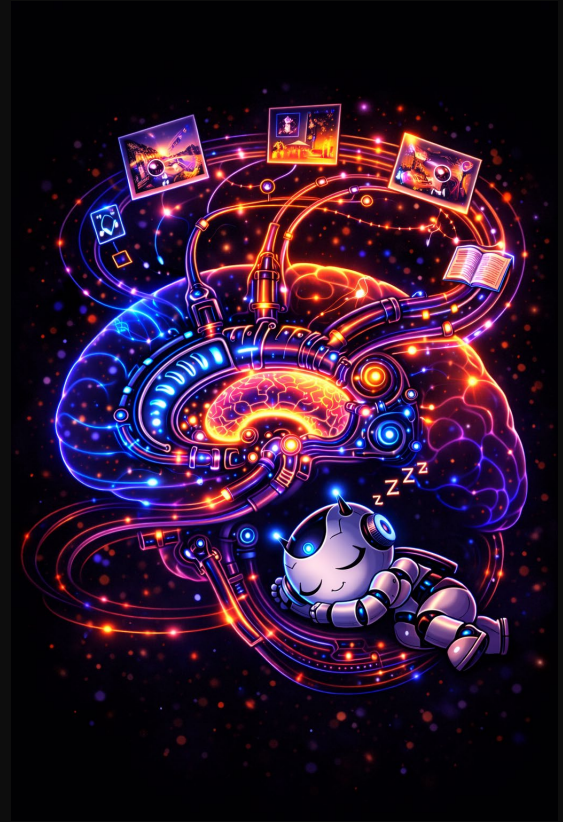
The Two-System Architecture:

Hippocampus: Fast learning, limited capacity, temporary

Cortex: Slow learning, vast capacity, permanent

During slow-wave sleep, the hippocampus REPLAYS recent experiences to the cortex — compressed and shuffled.

Skip sleep = hippocampus fills up, memories stay fragile





Experience Replay — From Hippocampus to DeepMind

Feature	Hippocampal Replay	DQN Experience Replay
When	During sleep	Between actions
What	Recent waking experiences	(state, action, reward, next state)
Order	Compressed, shuffled	Random from buffer
Purpose	Consolidate, prevent forgetting	Stabilize training
Priority	Emotional events replayed more	Prioritized by TD error

Demis Hassabis (DeepMind CEO) studied hippocampal memory in his PhD → directly inspired DQN (2013)

Prioritized experience replay (2016) mirrors emotional tagging — both systems prioritize surprising memories

Place Cells & Grid Cells — The Brain's GPS

Place cells (O'Keefe, 1971): Fire at specific locations

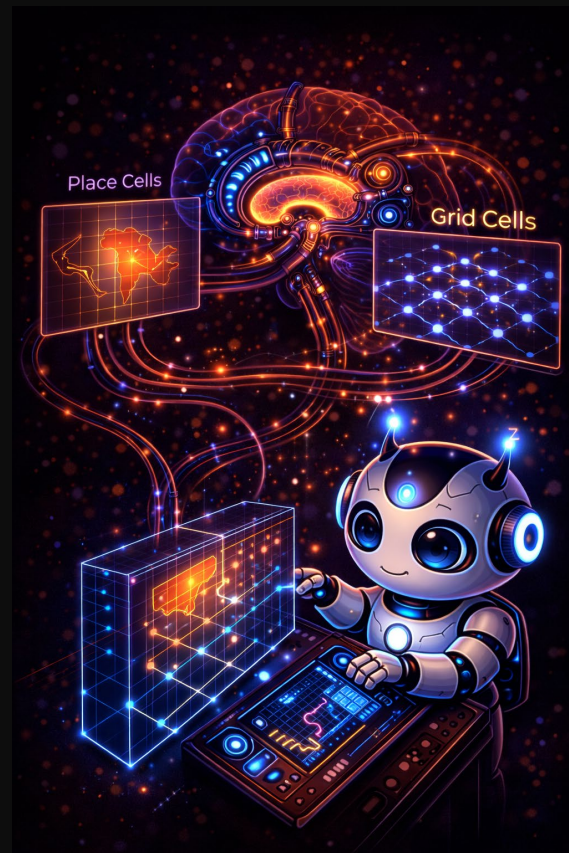
Grid cells (Moser, 2005): Fire in hexagonal patterns

Nobel Prize 2014

🧠 → 🤖 **Banino et al. (2018, Nature):**

AI navigation network spontaneously developed grid cell-like hexagonal patterns — without being designed to!

Convergent evolution: same problem → same solution





Complementary Learning Systems (CLS)

McClelland, McNaughton, & O'Reilly (1995)

The Problem: Catastrophic Forgetting

Train a network on cats → then train on dogs → it forgets cats entirely.

New learning overwrites old knowledge.

The Solution: Two Complementary Systems

Hippocampus quickly encodes new experiences without disturbing cortical knowledge

During sleep, replays experiences INTERLEAVED with old knowledge to the cortex

Cortex learns gradually from this mixed stream → no catastrophic overwriting

Knowledge Check — Memory Systems



1. Name the four memory systems and give a daily-life example of each.
2. Which system was destroyed in H.M.? Which system let him learn mirror tracing?
3. Why does the brain need TWO memory systems instead of one?
4. How did hippocampal replay inspire DeepMind's DQN?
5. A student pulls an all-nighter instead of sleeping. What happens to their learning?

LTP — Learning at the Synapse

Bliss & Lømo (1973) — the discovery

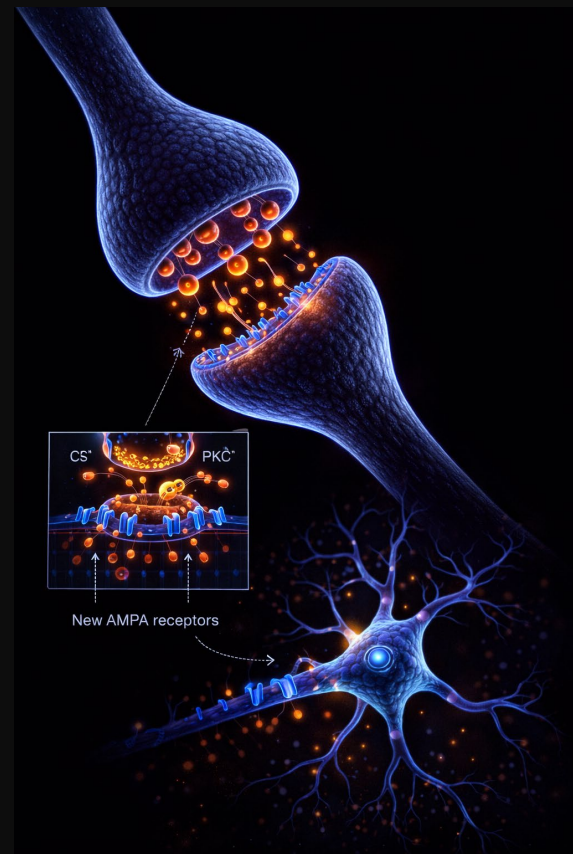
Rapid, repeated stimulation → connection becomes **permanently stronger**

The phone call analogy:

Two friends have an intense rapid-fire conversation → after that, they connect more easily and respond faster

NMDA receptor = coincidence detector

Only opens when BOTH neurons are active → Hebb's rule made real!



Long-Term Depression (LTD) The Brain's Eraser

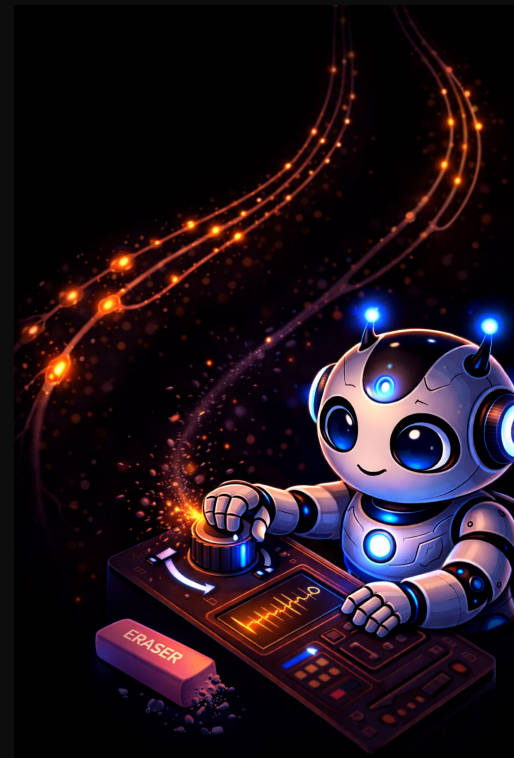
Slow, low-frequency stimulation → connection weakens

Together, LTP + LTD = a **bidirectional volume knob** for every connection

LTP (strengthen) = increasing a weight in AI | High-frequency stimulation

LTD (weaken) = decreasing a weight in AI | Low-frequency stimulation

Key difference: In AI, a central algorithm computes weight changes. In the brain, each synapse figures it out from local chemistry.



Chemical Gatekeepers — When to Learn

Neuromodulators don't cause neurons to fire — they control HOW MUCH the brain learns

Dopamine *"That was surprising!"*

Fires when outcome is BETTER than expected. Enhances LTP at active synapses.

🧠 → 🤖 Reward signal in RL (TD error)

Acetylcholine: *"Pay attention!"*

Released during alertness & focus. Turns up sensitivity to new input.

🧠 → 🤖 Attention-gated learning (open frontier)

Norepinephrine *"Something important!"*

Released during arousal, stress, novelty. Enhances memory across regions.

🧠 → 🤖 Explore vs. exploit (ϵ -greedy)

Key gap: AI uses ONE fixed learning rate. The brain uses MULTIPLE dynamic rates simultaneously.





Different Systems Learn at Different Speeds

System	Speed	Exposures?	Example
Episodic	ONE shot	Just one	Restaurant that gave you food poisoning
Semantic	Slow	Many over weeks	Building up history knowledge
Procedural	Very slow	Thousands of reps	Months of daily piano practice
Working	Instant (temp)	Zero — active	Holding a phone number while dialing

*Most AI systems have ONE learning rate for all parameters.
The brain spans ORDERS OF MAGNITUDE — from one-shot to thousands of reps.*

Not All Sleep Is Equal: Sleep Stages and Learning

Slow-Wave Sleep (Deep) — *First half of night*

Hippocampal replay to cortex

Consolidates: facts, events, textbook material

🧠 → 🤖 Experience replay in DRL

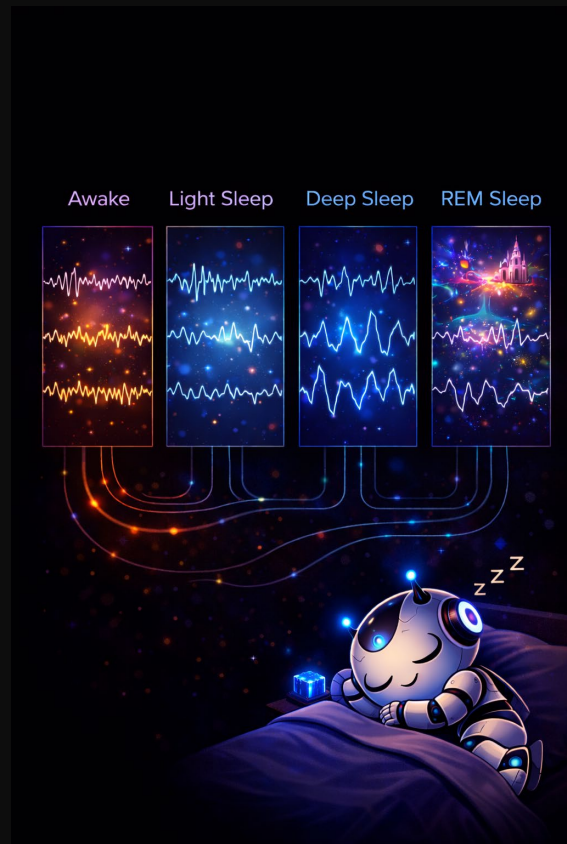
REM Sleep (Dreaming) — *Second half of night*

Cortical networks active, hippocampus quiet

Consolidates: motor skills, emotions, creativity

🧠 → 🤖 Generative replay, data augmentation

⚠️ **Skipping sleep doesn't give you more study time**
— it erases what you already studied.





Knowledge Check — How the Brain Learns

1. What is LTP? Explain it using the 'phone call' analogy.
2. Why does the brain need BOTH LTP and LTD?
3. Which neuromodulator signals 'that was surprising'? What does it do to learning?
4. A student practices piano, then studies history, then sleeps.
Which sleep stage helps each?

PART 2

How AI Remembers. And What It Gets Wrong.

Computational Parallels — With Optional Math Moments



LSTMs: The First AI Memory

Hochreiter & Schmidhuber (1997)

Problem: RNNs forget over long sequences
(vanishing gradients)

Solution: Three gates + a cell state highway

Input gate: what to store

Forget gate: what to discard

Output gate: what to use now

🧠 Bio parallel: Thalamic gating + active forgetting



Transformers: Memory Through Attention

Vaswani et al. (2017) — 'Attention Is All You Need'

Self-attention: every token can attend to every other token directly

No vanishing gradients — position 1 $\leftrightarrow$ position 1000 with equal strength

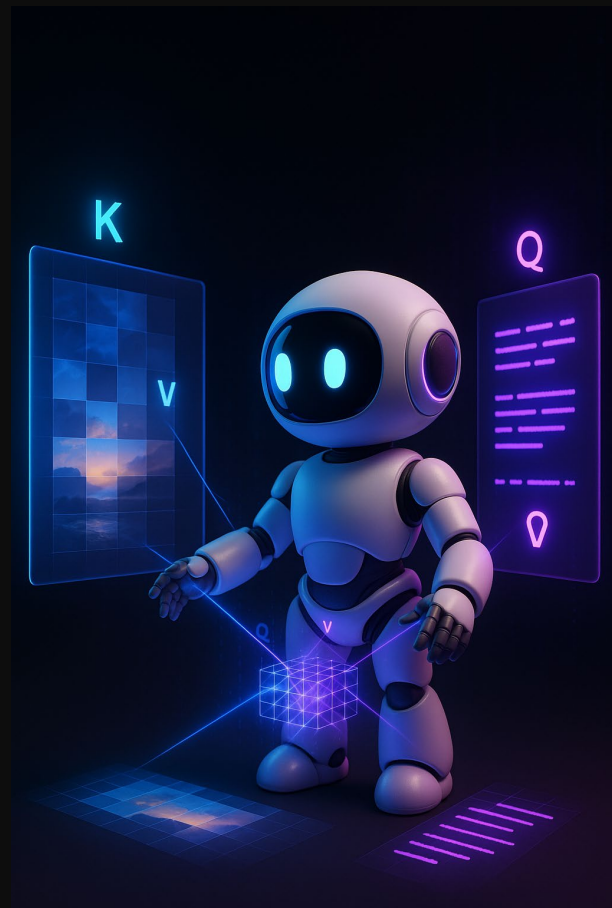
Query: 'What am I looking for?'

Key: 'What do I contain?'

Value: 'What info do I provide?'

🧠 **This IS content-addressable memory — just like the hippocampus!**

Partial cue (query) $\rightarrow$ retrieves stored patterns (key-value) based on similarity



External Memory — MANNs and RAG

Neural Turing Machine / DNC (*Graves et al., 2014/2016*)

Controller (PFC) + Memory matrix (hippocampus)

Read = pattern completion (CA3) | Write = encoding | Erase = active forgetting

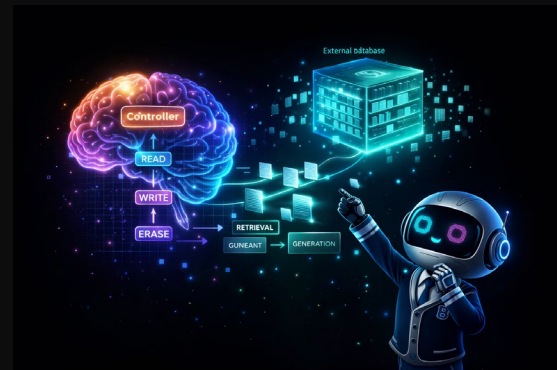
Learns to copy, sort, and navigate graphs

Retrieval-Augmented Generation (RAG) *The most practical AI memory today*

Model weights = cortical semantic memory

External DB = hippocampus (fast-updated, specific)

Query → retrieve relevant passages → generate



 **RAG is one of the most in-demand AI engineering skills right now**



Comparing AI Memory Architectures

Architecture	Memory Type	Capacity	Bio Parallel	Best For
RNN / LSTM	Internal (hidden state)	Small, fixed	Working memory	Short sequences
Transformer	Internal (context)	Medium, fixed	WM + attention	Text, code, reasoning
NTM / DNC	External (matrix)	Medium, flex	Hippocampal R/W	Algorithmic tasks
RAG	External (database)	Very large	Hippo-cortical	Knowledge QA
Exp. Replay	Buffer (FIFO)	Large	Hippo replay	Reinforcement learning



Knowledge Check — AI Memory

1. What problem do LSTMs solve? What mechanism do they use?
2. How is transformer self-attention like hippocampal memory retrieval?
3. How does RAG mirror the hippocampal-cortical distinction?
4. Which architecture for a chatbot that needs company documents?



Can the Brain Do Backpropagation?

Four reasons backprop is biologically implausible:

1 Weight Transport Problem

Backward pass needs exact copies of forward weights — no known bio mechanism

2 Non-local Error Signals

Neurons deep in the network adjust based on output error — but biology is LOCAL

3 Separate Forward/Backward Phases

Brain activity is continuous and bidirectional — no neat two-phase computation

4 Precise Real-Valued Derivatives

Neurons use all-or-nothing spikes — not precise floating-point numbers

Hebbian Learning: 'Fire Together, Wire Together'

Donald Hebb (1949) — the brain's oldest learning rule

Two neurons active together → connection strengthens
Out of sync → connection weakens

STDP (Module 04) is the precise timing version of this rule
LTP is the cellular mechanism that makes it work

✓ **Biologically plausible**

✗ **Cannot train deep networks (no error signal)**





Bridging the Gap — Bio-Plausible Alternatives

Feedback Alignment *(Lillicrap 2016)*

Random fixed feedback weights instead of weight transport. Works ~95% as well!

Predictive Coding *(Rao & Ballard 1999)*

Each layer predicts the layer below; only errors propagate. Local computation = equivalent to backprop!

Forward-Forward *(Hinton 2022)*

Two forward passes (positive + negative data) — no backward pass at all.

Reward-Modulated STDP

STDP + dopamine reward signal. Eligibility traces solve credit assignment biologically.

Predictive coding connects to Module 02 (Free Energy Principle) — the brain as a prediction machine



Bio-Plausibility Scorecard

Algorithm	Weight Transport?	Global Error?	Separate Phases?	vs. Backprop
Backpropagation	Yes ✗	Yes ✗	Yes ✗	Gold standard
Hebbian / STDP	No ✓	No ✓	No ✓	Limited
Feedback Alignment	No ✓	Partial	Yes	~95%
Predictive Coding	No ✓	Local ✓	No ✓	~Equivalent
Forward-Forward	No ✓	No ✓	No ✓	~90%

✓ Knowledge Check — Biological Plausibility



1. Name two reasons backprop is biologically implausible.
2. Why is Hebbian learning plausible but limited?
3. How does predictive coding connect to Module 02's prediction machine?
4. For a neuromorphic chip that learns on-device, which algorithm would you choose?



The Big Picture — Biology → AI Status

Biological Principle	Status in AI
Multiple memory systems	Partially adopted
Fast encoding + slow consolidation	Adopted (experience replay)
Content-addressable retrieval	Adopted (attention, RAG)
Pattern separation	Rarely used
Emotional tagging	Partial (prioritized replay)
Active forgetting	Emerging (EWC, pruning)
Dynamic learning rates (neuromodulators)	Mostly absent
Biologically plausible learning	Active research

Biggest gap: FLEXIBILITY — context-sensitive, emotionally modulated, continuously updating, reconstructive memory



Module 06 — Key Takeaways

- 1 Memory is four systems (working, episodic, semantic, procedural) with separate brain structures and AI parallels
- 2 Two-system architecture (hippocampus + cortex) solves the fast-learning vs. stable-storage tradeoff
- 3 LTP/LTD + neuromodulators = the brain's dynamic, context-dependent learning machinery
- 4 Experience replay: clearest neuroscience → AI success story. Sleep is part of learning.
- 5 Predictive coding may let the brain do backprop-equivalent learning with local biology
- 6 Biggest gap: AI memory lacks the flexibility, emotional modulation, and continuous updating of biological memory



Next: Module 07 Attention & Consciousness in AI

We've covered: SENSE → REMEMBER & LEARN

Next question: How does the brain DECIDE what to focus on?

Coming in Module 07:

Biological attention: spatial, feature-based, executive

Global Workspace Theory & Integrated Information Theory

Attention mechanisms in AI vs. biological attention

The hard problem of consciousness — and whether AI needs it

Teaser: Transformers broadcast information globally, just like consciousness theories say the brain does. Does that make them conscious?



Discussion

If you could give an AI system ONE biological memory feature it currently lacks, which would you choose and why?
